

POLITECNICO
MILANO 1863

# Best Bike Paths – DD Document

DD document for the
Software Engineering 2 course assignment

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# 1 Introduction

## 1.1. Purpose

The purpose of this document is to serve as a detailed descriptor of the BBP project, to be used by actual developers and as an actual agreement between customer and contractors: it will contain detailed informations about the architecture of the system, its full functionalities, constraints and it will allow the customer to validate the integrity and completeness of said system. With this DD, the customer will also be able to validate all of the requirements, to verify if the system finally meets the expectations.

## 1.2. Scope

The Best Bike Paths (BBP) project, initiated by a leading bike association, aims to be an advanced support system for the cycling community designed to thrive within a transport landscape dominated by cars.

The core idea is to foster a collaborative digital environment where cyclists can actively record, consult, and publish both their own and other users' riding paths.

This platform will serve as the definitive centralized resource for finding highly recommended routes tailored precisely to the user's specific experience, preferences, and safety considerations.

Ultimately, the system aims to significantly enhance the quality of the cycling experience by providing verified, community-rated routes, thereby connecting riders and making high-quality path discovery universally accessible

## 1.3. Definitions, Acronyms, Abbreviations

- BBP: Best Bike Paths
- RASD: Requirements Analysis and Specification Document
- DD: Design Document
- IIT: Implementation, Integration and Test Plan

- WAF: Web Application Firewall

## 1.4. Reference Documents

- Specification document assignment
- IEEE standard documentation for DD
- Materials for the Software Engineering 2 course made available on WeBeep

## 1.5. Document Structure

The document is organized in seven different chapters that will cover every detail on the actual design of our project:

- **Chapter 1 - Introduction:** this chapter shows the aim of the doucment, what it is needed to know before to actually start the read of the whole text and restates the scope of the project.
- **Chapter 2 - Architectural Design:** this chapter shows all of the main components of the whole system, including architectural choices and patterns used.
- **Chapter 3 - User Interface Design:** this chapter shows and explains how the mockup pages we created integrate all of the implemented functionalities.
- **Chapter 4 - Requirements Traceability:** this chapter shows how we ensured that every requirement needed to the system to work is being satisfied with relative design choices.
- **Chapter 5 - Implementation, Integration and Test Plan:** this chapter shows how all of the components are implemented and their integration inside the system, also providing a test plan for all of their instances.
- **Chapter 6 - Effort Spent:** this chapter includes a table where the amount of hour of commitment of every participant is shown relative to its section.
- **Chapter 7 - References:** finally, the last chapter contains all of our document references used to better refine our DD.

# 2 Architectural Design

## 2.1. Overview

## 2.2. Component View

### 2.2.1. Mobile

### 2.2.2. Backend

## 2.3. Deployment View

In this section of the document we will be focusing on the whole deployment of BBP and the motivations behind our choices.

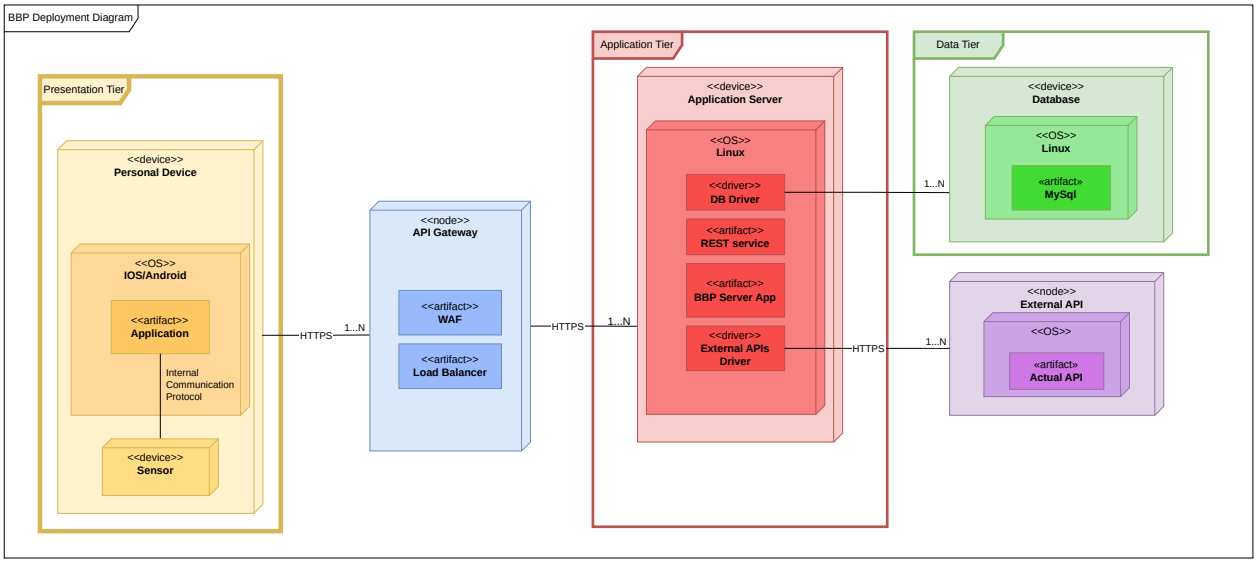



Figure 2.1: Deployment View

We divided our system into three main tiers: presentation, application and data tiers (better explained in Section 2.6) to better isolate common features between all tangible and intangible instances of all the objects we used. Below, a list of all the actors and objects involved:

- **Personal Device**: a personal device consists of a Sensor device (here modeled as the

union of all the device's movement sensors) and an OS that can be either Android or iOS. It is to be intended as a personal piece of hardware such as a smartphone or smartwatch that is carried by the user throughout the execution of the application. The personal device communicates with the Application Server using a local view instance of the app, here called Application: every request made by the Application to the Application Server follows a RESTful interaction style over HTTPS.

- **API Gateway**: it's a node offered by a third party (such as AWS, Cloudflare, Azure) that has a built-in WAF and load balancer to better distribute traffic to all the distributed Application Servers avoiding single points of failure.
- **Application Server**: it's a device that contains the logic of the application and connects together all the drivers needed to the application to work. It is connected with external APIs using REST and with multiple databases using a continuously working connection pool.
- **Database**: we decided to have different databases (one main database, more replicas used only in read mode) with a connection pool contained inside the Application Server, so that databases are actually protected by a limited amount of opened connection and overload is not possible. This feature is extremely useful to minimize concurrency problems and the number of connections can be adjusted accordingly to the number of requests seen to have the optimal balance of hardware usage and system efficiency: we chose Linux as the OS because of its open source roots and MySQL for its robustness.

Now for a closer look at some of the components of the devices themselves:

- **Application Artifact**: the client side application. It is used primary as the presentation layer and stores a signed JWT (JSON Web Token) used for authentication: it is connected with all sensors needed to detect movements in automated mode using cross-platform mobile frameworks.
- **BBP Server App**: the actual model of our application. Here, the logic of the app is stored and ensures all operations are coherent with user requests following the MVC pattern: it is the recipient of all the managers and connects them to every driver needed to interface to external tools.

## 2.4. Runtime View

## 2.5. Component Interfaces

## 2.6. Styles and Patterns

## 2.7. Other Design Decisions

# 3 | User Interface Design

# 4 | Requirements Traceability

# 5 Implementation, Integration and Test Plan

## 5.1. Overview and Implementation Plan

## 5.2. Features identification

## 5.3. Integration Strategy

## 5.4. System Testing Strategy

# 6 Effort Spent

A table representing the total hours of work is shown here.

| Work Areas | Alessandro Burato | Gabriele Crimella | Davide Faccini |
|---|---|---|---|
| Introduction | 1h | 0h | 0h |
| Architectural Design | 8h | 0h | 0h |
| UI Design | 0h | 0h | 0h |
| Requirements Traceability | 0h | 0h | 0h |
| IIT | 0h | 0h | 0h |
| Group Meetings and Reviews | 0h | 0h | 0h |

Table 6.1: Table of Effort

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